PSYCHOMETRIC INSTABILITY OF LLM BENCHMARKS: DIFFERENTIAL ITEM FUNCTIONING REVEALS 31% OF MMLU ITEMS BEHAVE DIFFERENTLY ACROSS MODEL GENERATIONS

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ABSTRACT

LLM benchmark scores are routinely compared across model generations, yet the assumption that individual items measure the same construct over time is rarely tested. We apply Mantel-Haenszel Differential Item Functioning (DIF) analysis with ETS severity classification to the METABENCH response matrix (5,227 models $\times$ 12,508 MMLU items), and find that 31% of items exhibit large DIF (ETS Category C) between 2023 and 2024 model cohorts. This instability does not threaten benchmark validity: Spearman rank correlation between full and stable-item rankings is $\rho \approx 0.997$. Rather, it reveals which capability dimensions shift across model generations. Item-level decomposition shows that domain accuracy improvement explains only 2.4 percentage points (7.6%) of the instability; 92.4% is item-specific. We expose a validation methodology gap in the contamination detection literature: web-overlap labels measure global exposure, not differential functioning, and the enrichment between DIF flags and these labels is 0.96 (chance level). A semi-synthetic injection framework with cross-subject matching achieves $\Delta\mathrm{FPR} = 0.000$ while preserving detection power ($\Delta\mathrm{TPR} = +0.508$ at exploitation rate $p = 0.5$); cross-benchmark replication on GSM8K confirms generality ($\Delta\mathrm{TPR} = +0.818$). DIF-flagged items concentrate cross-generational discriminative information ($z = -8$ to -49σ vs. random removal) and anti-correlate with the GSM1K contamination gap ($r = -0.478$, $p = 0.005$, $N = 33$), suggesting that DIF captures genuine capability change rather than memorization. LLM benchmarks need psychometric monitoring, not just score reporting.

1 INTRODUCTION

When we say “Model B outperforms Model A on MMLU,” we implicitly assume that the benchmark measures the same construct in the same way for both models. This assumption of temporal comparability underpins every cross-generation comparison published on leaderboards and in model reports, yet it is rarely tested. Applying Mantel-Haenszel Differential Item Functioning (MH-DIF) analysis (Holland & Thayer, 1988) with Educational Testing Service (ETS) severity classification to the METABENCH response matrix (Kipnis et al., 2024), we find that 31% of MMLU (Hendrycks et al., 2021) items receive ETS Category C classification ($|\Delta_{\mathrm{MH}}| \geq 1.5$, $p < 0.05$), indicating large shifts in item functioning between models released in 2023 and 2024. This is not statistical noise: a random-split control within the 2023 cohort yields only 0.18% ETS C items. The instability persists within every MMLU subject domain, with domain-level C% ranging from 31% to 45% (mean 38.1%), and item-level regression shows that 92.4% of this instability is unrelated to aggregate domain-level accuracy gains. Nearly one in three MMLU items behaves differently depending on when the evaluated models were trained. Crucially, this instability does not threaten benchmark validity: Spearman rank correlation between model rankings computed on all items versus stable-only items is $\rho \approx 0.997$. The instability is diagnostic, not destructive; it reveals which capability dimensions shift across model generations rather than rendering MMLU scores unreliable.

A growing body of contamination detection research has produced sophisticated item-level methods, from membership inference approaches such as Min-K% (Shi et al., 2024) to performance-based

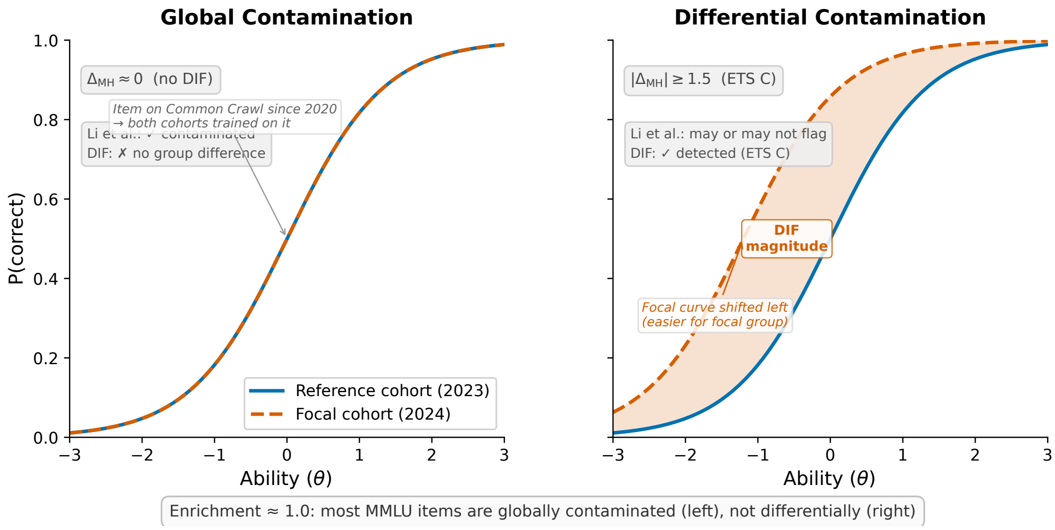

Figure 1: Global vs. differential contamination, illustrated via Item Characteristic Curves (ICCs). **Left**: global contamination produces no DIF; both cohorts encounter the item, and their ICCs overlap. **Right**: differential contamination produces DIF; the focal cohort's ICC shifts because only that cohort was disproportionately exposed during training. The enrichment ratio of ≈ 1.0 between DIF flags and web-overlap labels is expected because most MMLU items fall into the left category.

detectors such as ConStat (Dekoninck et al., 2024), DVD (Liang, 2026), and Self-Critique (Tao, 2026). These methods validate detection accuracy primarily against web-overlap labels such as those of Li et al. (2024b), which classify benchmark items as contaminated based on verbatim or near-verbatim matches in publicly accessible training corpora. Such labels measure *global exposure*: whether an item exists in the data ecosystem that all training pipelines can access. They do not measure *differential item functioning*: whether an item behaves differently across model cohorts that may have had varying degrees of exposure or different training methodologies. Figure 1 illustrates the distinction. Most MMLU items are globally contaminated in the sense that they exist in web-accessible text, so all model cohorts encounter them and no differential signal arises. The items that function differentially are those where exposure or training methodology changed asymmetrically across cohorts. Validating differential detectors against global-exposure labels therefore conflates two phenomena, and the near-chance enrichment (0.96) between DIF flags and web-overlap labels is the expected outcome of this mismatch.

We apply classical MH-DIF with ETS classification to the full METABENCH response matrix (5,227 models $\times$ 12,508 MMLU items), the largest-scale DIF analysis on LLM benchmarks to date. This psychometric approach complements concurrent work on anchor-item calibration for score comparability (Habba et al., 2026) and addresses the fragility of existing contamination detectors under post-training perturbation (Wang et al., 2026). Semi-synthetic injection experiments confirm that DIF detects differential contamination when present ($\Delta\text{TPR} = +0.508$ at exploitation rate $p = 0.5$), and our cross-subject matching protocol eliminates false positive inflation ($\Delta\text{FPR} = 0.000$). Three structural controls (matched-ability quintile analysis, IRT θ-conditioning, and within-subject pre-checks) confirm that the 31% temporal instability cannot be attributed to methodological artifacts, establishing it as a robust empirical finding about MMLU's measurement properties.

MMLU's 12,508 items provide sufficient measurement redundancy that removing nearly a third does not alter rank order. DIF should therefore be understood as a diagnostic lens (which items changed and for which model families) rather than a quality filter (remove bad items to improve the benchmark). Despite aggregate ranking stability, DIF-flagged items concentrate discriminative signal that differentiates model generations: removing DIF items degrades ranking discriminability more than removing an equal number of random items ($z = -8$ to -49σ across 21 conditions). Individual model families show distinctive DIF profiles: Phi-3 and Llama-3 gain $+700$ to $+850$ rank positions under stable-item evaluation, while Mistral-based merges lose -400 to -587 positions.

We do not modify the MH-DIF procedure itself; the contributions are empirical, not algorithmic. Three layers of genuine novelty emerge: (a) we discover that 31% temporal instability is robust across thresholds and unrelated to capability improvement, the first large-scale characterization of item-level benchmark stability at 5,227 models; (b) we expose a fundamental validation methodology gap in the contamination detection literature, showing that web-overlap labels cannot serve as ground truth for item-level differential detectors; and (c) we provide a reusable semi-synthetic validation framework with cross-subject matching ($\Delta\mathrm{FPR} = 0.000$), confirmed on GSM8K with independent contamination validation through the GSM1K held-out set (Zhang, 2024). We make the following contributions:

1. **Benchmark temporal comparability** (Claim 1): 31% of MMLU items are functionally unstable across model generations under ETS C classification, with the instability persisting within every subject domain (mean C% $= 38.1\%$) and confirmed non-random by random-split control at 0.18% (Section 4.2).
2. **Ground truth meta-critique** (Claim 2): Web-overlap contamination labels (Li et al., 2024b) cannot evaluate item-level differential detectors. The enrichment between DIF flags and web-overlap labels is 0.96 (chance level), exposing a validation methodology gap in the contamination detection literature (Section 4.3).
3. **Semi-synthetic validation framework** (Claim 3): Controlled injection with cross-subject matching achieves $\Delta\mathrm{FPR} = 0.000$ while preserving detection power ($\Delta\mathrm{TPR} = +0.508$ at $p = 0.5$); cross-benchmark replication on GSM8K confirms generality (Sections 4.5 and 4.7).
4. **DIF as diagnostic lens** (Claim 4): DIF-flagged items concentrate discriminative information ($z = -8$ to -49σ vs. random removal) and anti-correlate with the GSM1K contamination gap ($r = -0.478$, suggestive at $N = 33$), indicating that DIF captures item-level behavioral change distinct from simple memorization (Sections 4.6 and 4.7).

2 Related Work

Contamination Detection. A growing family of methods detects benchmark contamination through model behavior or training data analysis. Membership inference approaches estimate whether a model has seen specific items during training: Min-K% (Shi et al., 2024) and Min-K%++ (Zhang et al., 2024) use token-level log-likelihood statistics, while Time Travel (Golchin & Surdeanu, 2024) probes models with temporal cues. Performance-based detectors identify contamination through statistical anomalies: ConStat (Dekoninck et al., 2024) tests performance gaps between original and perturbed items, DVD (Liang, 2026) uses distributional verification, and Self-Critique (Tao, 2026) employs model self-evaluation. Balloccu et al. (2024) provide a broader taxonomy of leakage detection strategies. These methods share a common validation practice: they verify detection accuracy against web-overlap or n-gram labels (Li et al., 2024b) that classify items based on verbatim matches in publicly accessible corpora. Such labels measure *global exposure* (whether an item exists in training data) rather than *differential item functioning* (whether an item behaves differently across model cohorts). This distinction is central to our Claim 2 (Section 4.3): web-overlap labels cannot serve as ground truth for differential detectors, because most benchmark items are globally exposed to all model cohorts. Wang et al. (2026) show that existing contamination detectors are fragile under reinforcement learning, which further motivates behavioral approaches grounded in psychometric theory rather than NLL-based signals.

IRT in LLM Evaluation. Item Response Theory (IRT) provides a principled framework for jointly analyzing test items and examinee ability, and its adoption in LLM evaluation has accelerated. Lalor et al. (2019) introduced IRT to NLP by fitting item parameters to human and model response data; their analysis revealed that item difficulty and discrimination vary substantially across benchmark questions. Vania et al. (2021) applied IRT to assess test set discriminability and found that many items in standard NLP benchmarks contribute little information about model differences. Recent extensions include PSN-IRT (Zhou et al., 2026), which fits four-parameter logistic models to characterize benchmark quality at scale, and scalability coefficients (Hardy et al., 2026) as lightweight item quality pre-filters. TinyBenchmarks (Maia Polo et al., 2024) uses IRT-based item selection to construct representative benchmark subsets that predict full-benchmark scores with far fewer items. ATLAS

(Li et al., 2025) and IRTNet (Chen et al., 2025) explore neural extensions of classical IRT for LLM ability estimation, and Choi et al. (2026) apply IRT to calibrate LLM-as-judge evaluations.

The closest concurrent work is Growing Pains (Habba et al., 2026), which addresses score comparability across model generations through fixed-parameter calibration (FPC; Kim & Kolen, 2019). FPC anchors a common ability scale by fixing the item parameters of a shared anchor set, so that models calibrated at different times receive scores on the same metric. Their framework selects 100 anchor items with stable IRT parameters and calibrates new models against this fixed scale, which enables extensible evaluation without full re-calibration. Our work is complementary: Growing Pains assumes anchor item stability, while DIF analysis diagnoses which items violate this assumption. The combination yields a monitored extensible evaluation system: DIF flags anchor items whose functioning has degraded, and the calibration framework triggers re-selection when the stable anchor set shrinks below a validity threshold.

DIF Methodology and Measurement Invariance. The Mantel-Haenszel procedure for DIF detection was introduced by Holland & Thayer (1988) and remains the most widely used method in educational testing due to its simplicity and well-characterized statistical properties (Clauser & Mazor, 1998). Alternative approaches include logistic regression DIF (Swaminathan & Rogers, 1990), SIBTEST (Shealy & Stout, 1993), Lord's χ^2 test (Lord, 1980), and the Raju area method (Raju, 1988); Zumbo (1999) provides a comprehensive comparison. DIF analysis is a specific instance of measurement invariance testing (Meredith, 1993; Vandenberg & Lance, 2000), which examines whether a measurement instrument functions equivalently across populations. MIMIC models offer a latent-variable approach to the same problem (Muthén, 1985; Woods, 2009), and Teresi & Fleishman (2007) reviews DIF methods across applied domains. We do not modify the MH procedure; our contribution is the application context. The scale of this analysis (5,227 examinees across a 57-subject multidimensional ability space) exceeds typical educational testing applications, where Belzak (2020) note that DIF methods can behave differently at extreme sample sizes. The primary methodological challenge is the impact-versus-DIF confound: genuine ability differences between temporal cohorts can inflate DIF estimates. We address this through matched-ability quintile analysis, IRT θ-conditioning, and a random-split control (Section 4.2).

Benchmark Design and Concurrent Work. Our analysis psychometrically characterizes an existing benchmark (MMLU; Hendrycks et al., 2021) rather than constructing a new one. Several concurrent efforts address benchmark limitations through design. Dynamic benchmarks such as Dynabench (Kiela et al., 2021) and LatestEval (Li et al., 2024a) generate items adversarially or from recent sources to prevent memorization, and LiveBench (White, 2025) refreshes items monthly from contemporary material. Chatbot Arena (Chiang et al., 2024) and GAIA (Mialon et al., 2023) sidestep static benchmarks entirely through pairwise human preference or real-world task completion. HELM (Liang et al., 2023) standardizes evaluation across tasks but does not address item-level stability. These approaches mitigate contamination by construction, yet none provides psychometric equating across versions: a model's score on LiveBench v3 is not directly comparable to its score on v1 without an anchoring mechanism such as that of Habba et al. (2026). METABENCH (Kipnis et al., 2024) provides the large-scale response matrix (5,227 models $\times$ 12,508 items) that makes our analysis possible.

LLM Evaluation Methodology. Broader surveys of LLM evaluation (Chang et al., 2024) identify benchmark saturation and lack of item-level transparency as systemic challenges. Burnell et al. (2023) argue that current practices treat benchmarks as monolithic scores, which obscures item-level variation in difficulty, discrimination, and stability. MMLU-Pro (Wang et al., 2024) addresses saturation by increasing item difficulty, and Ott et al. (2022) document how benchmark scores plateau as models improve. These diagnoses converge on a common gap: the absence of tools that operate at the item level rather than the benchmark level. Our DIF analysis fills this role by identifying which items within a benchmark have become unreliable for cross-generation comparison.

3 METHOD

We adopt Differential Item Functioning (DIF) analysis from educational measurement to assess whether individual benchmark items behave consistently across groups of LLMs. The procedure requires only a binary response matrix and group labels, with no access to model internals or training data. Below we define the data, the statistical test, the cohort designs, a semi-synthetic validation framework, diagnostic checks, and statistical controls.

3.1 Data

The primary dataset is the METABENCH response matrix (Kipnis et al., 2024), which records binary outcomes (1 = correct, 0 = incorrect) for 5,227 models evaluated on 12,508 MMLU items (Hendrycks et al., 2021). Models span release dates from 2023 to 2024, drawn from the Open LLM Leaderboard. MMLU covers 57 subjects grouped into five broad domains (STEM, Humanities, Social Sciences, Other, and a mixed category). We supplement the response matrix with the web-overlap contamination labels of Li et al. (2024b), which classify 9,229 of 12,508 items (73.8% coverage) as "contaminated" or "clean" based on verbatim matches in publicly accessible web corpora. These labels serve as an external reference for evaluating the relationship between DIF flags and existing contamination indicators.

3.2 Mantel-Haenszel DIF and ETS Classification

In educational testing, Differential Item Functioning (DIF) occurs when an item's statistical relationship to the latent trait θ differs between two groups after conditioning on ability level. An item with DIF is not merely harder for one group; it functions differently relative to what the group's overall ability would predict. The standard terminology refers to test-takers as "examinees" and questions as "items." In our setting, examinees are LLMs and items are benchmark questions.

The Mantel-Haenszel (MH) procedure (Holland & Thayer, 1988) detects DIF by stratifying examinees into K score bins based on total score (a proxy for θ), where K equals the number of distinct total scores observed in the combined sample, and computing a common odds ratio across strata. Within each stratum k, the responses of the reference group and the focal group form a 2×2 contingency table:

$$\alpha_{\mathrm{MH}} = \frac{\displaystyle\sum_{k=1}^{K} A_k D_k \,/\, T_k}{\displaystyle\sum_{k=1}^{K} B_k C_k \,/\, T_k}\,, \tag{1}$$

where A_k and B_k are the number of correct and incorrect responses in the reference group for stratum k, C_k and D_k are the corresponding counts for the focal group, and T_k is the total number of examinees in stratum k. Under the null hypothesis of no DIF, $\alpha_{\mathrm{MH}} = 1$. The log odds ratio is converted to the ETS delta scale:

$$\Delta_{\mathrm{MH}} = -2.35 \ln(\alpha_{\mathrm{MH}})\,. \tag{2}$$

The Educational Testing Service (ETS) classifies DIF severity into three categories based on $|\Delta_{\mathrm{MH}}|$. Category A (negligible) requires $|\Delta_{\mathrm{MH}}| < 1.0$. Category C (large) requires both $|\Delta_{\mathrm{MH}}| \geq 1.5$ and statistical significance at $p < 0.05$ from the MH chi-squared test. Category B (moderate) includes all remaining items: those with $1.0 \leq |\Delta_{\mathrm{MH}}| < 1.5$, or $|\Delta_{\mathrm{MH}}| \geq 1.5$ without statistical significance. The dual threshold in ETS C (effect size *and* significance) guards against flagging items that show large but unreliable effect sizes. Throughout this paper, we report the percentage of items classified as ETS C, denoted C%.

The choice of matching variable (the total score used for stratification) affects both validity and interpretation. We distinguish three matching strategies: (1) *standard matching* uses the within-benchmark total score on all items of the same benchmark; (2) *cross-subject matching* uses the total score on all subjects *other than* the one containing the item under test, preventing the matching variable from absorbing injected contamination signals (Section 3.4); and (3) *external matching* uses the total score on a different benchmark entirely (e.g., MMLU total score as the matching variable when testing DIF on GSM8K items), providing an independent ability proxy uncontaminated by the target benchmark. Cross-subject and external matching both address matching variable contamination but at different granularities: cross-subject operates within a multi-domain benchmark, while external operates across benchmarks.

3.3 Cohort Designs

DIF requires defining a reference group and a focal group. We construct five cohort comparisons, each designed to isolate a different source of variation.

Temporal. The reference group consists of models released in 2023 ($N = 1{,}080$) and the focal group consists of models released in 2024 ($N = 807$). This is the primary comparison for detecting items whose functioning changed across model generations.

Ability-quintile. Models within the same temporal window are divided by total score into quintiles. The focal group contains the lowest-scoring models (Q1–Q2) and the reference group contains the highest-scoring models (Q4–Q5). This comparison tests whether ability differences alone produce DIF, without temporal confounds.

Within-family. Llama-2 models ($N = 161$) serve as the reference group and Llama-3 models ($N = 308$) as the focal group. This comparison isolates generational changes within a single model family.

Within-subject. DIF is computed separately for each of the 57 MMLU subjects, with the matching variable (total score) restricted to items within the same subject. This design tests whether instability concentrates in specific knowledge domains.

Random split. The 2023 cohort is divided into two equal random halves ($N = 540$ each). This serves as a sanity check: no systematic DIF should appear between two random subsets of the same population (expected C% ≈ 0).

3.4 Semi-Synthetic Injection Framework

To evaluate whether MH-DIF can detect differential contamination when it is present, we construct a semi-synthetic framework with controlled ground truth.

Procedure. From items classified as ETS A in the uninjected data (negligible DIF), we select 20% uniformly at random as the "contaminated" set. For each contaminated item, we flip incorrect responses to correct in the focal group at exploitation rate $p \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$. We then run MH-DIF on the modified response matrix and measure detection performance. We report $\Delta\mathrm{TPR} = \mathrm{TPR}(\text{injected}) - \mathrm{TPR}(\text{baseline})$ and $\Delta\mathrm{FPR} = \mathrm{FPR}(\text{injected}) - \mathrm{FPR}(\text{baseline})$, where TPR is the true positive rate on injected items and FPR is the false positive rate on uninjected items. Each injection level is repeated across 5 random seeds for uncertainty estimation.

Cross-subject matching. The choice of matching variable is critical for valid semi-synthetic evaluation. When contamination is injected into items within subject X, the total score on subject X absorbs part of the injection signal. Using this contaminated total score as the MH matching variable violates the conditional independence assumption and inflates FPR. Cross-subject matching resolves this problem by computing the matching variable from all subjects *other than* the one being tested. Under this protocol, $\Delta\mathrm{FPR} = 0.000$ across all five MMLU domains, while detection power is preserved ($\Delta\mathrm{TPR} > 0.3$ at $p = 0.5$ for four of five subjects). We adopt cross-subject matching as the standard protocol for all semi-synthetic analyses.

3.5 Diagnostics

Three diagnostic procedures complement the primary DIF analysis.

Local independence (Yen's Q3). The MH procedure assumes that item responses are conditionally independent given θ. Yen's Q3 statistic measures residual correlations between item pairs after conditioning on the latent trait. We compute Q3 for all within-subject and cross-subject item pairs, with $|Q3| > 0.20$ as the threshold for local dependence. Elevated within-subject dependence would indicate that items within the same subject share variance beyond what θ captures, which could inflate within-subject DIF rates.

Domain-specificity. We test whether temporal DIF concentrates in domains where models improved most. At the ecological level, we compute Spearman ρ between subject-level C% and subject-level accuracy improvement from 2023 to 2024. At the item level, logistic regression predicts item-level DIF C classification from domain-level accuracy change, with and without item property controls (difficulty, discrimination). This decomposition quantifies how much of the observed instability is attributable to aggregate domain-level gains versus item-specific behavioral change.

Instability decomposition. When domain accuracy improvement is included alongside item-level controls, we compute the counterfactual C% that would remain after removing domain-level effects. The gap between observed and counterfactual C% measures the contribution of domain-level improvement to the total instability.

3.6 Statistical Considerations

Multiple testing. The ETS C classification imposes a dual threshold (effect size $|\Delta_{\mathrm{MH}}| \geq 1.5$ and significance $p < 0.05$), which provides implicit control over false positives. As a robustness check, we apply Benjamini-Hochberg FDR correction at $q = 0.05$. The random-split sanity check provides an empirical estimate of the false positive rate under the null hypothesis.

Threshold sensitivity. We report results across a range of DIF thresholds ($|\Delta_{\mathrm{MH}}| \in \{1.0, 1.5, 2.0, 2.5\}$) to verify that qualitative conclusions do not depend on the specific ETS C cutoff. The standard threshold of $|\Delta_{\mathrm{MH}}| \geq 1.5$ corresponds to the ETS C definition used throughout the paper.

Architecture heterogeneity. The METABENCH MMLU population consists of 99.98% decoder-only models (5,226 of 5,227). This near-complete architectural homogeneity eliminates architecture as a confound for DIF analysis. We verify this by excluding the single encoder-decoder model and confirming identical results.

4 Experiments

We apply MH-DIF to the METABENCH dataset (Kipnis et al., 2024), which provides sparse response matrices for 5,227 models evaluated on 12,508 MMLU items (Hendrycks et al., 2021). All temporal analyses define reference and focal cohorts by model release date (2023 vs. 2024). We validate detection power through simulation, characterize temporal instability across multiple cohort designs, evaluate the relationship between DIF flags and existing contamination labels, and assess downstream consequences for benchmark rankings. Replication on GSM8K with cross-benchmark contamination validation appears in Section 4.7.

4.1 Validation: DIF Detection Power

To confirm that MH-DIF detects differential item functioning at the sample sizes available in METABENCH, we conduct a power analysis using empirical MMLU IRT parameters. We simulate item response matrices where a controlled subset of items has injected DIF of varying magnitude ($|\Delta_{\mathrm{MH}}| \geq 1.4$) and evaluate ETS C classification across sample sizes $N \in \{40, 80, 160, 320\}$ per cohort. At $N \geq 80$, ETS C achieves true positive rates of 0.68–0.80 with false positive rates of 0.036–0.042 and AUC of 0.93–0.94. The temporal cohort sizes in our analysis ($N_{\mathrm{ref}} = 1{,}080$, $N_{\mathrm{focal}} = 807$) far exceed this threshold, and statistical power is not a limiting factor for the results reported below. Full simulation results with confidence intervals appear in Appendix Table A1.

Table 1: DIF landscape across cohort definitions. ETS C denotes the percentage of items with $|\Delta_{\mathrm{MH}}| \geq 1.5$. FPR is measured against Li et al. web-overlap labels on the "clean" (non-overlapping) subset. Enrichment is the ratio of observed to expected overlap between DIF flags and web-overlap labels.

Cohort Definition	N_{ref} / N_{focal}	ETS C (%)	FPR (Li clean)	Enrichment
Temporal (2023 vs. 2024)	1,080 / 807	31.3	0.318	0.96
Ability quintile (Q1–2 vs. Q4–5)	∼2,600 / ∼2,600	2.9	0.027	1.19
Within-family (Llama-2 vs. 3)	161 / 308	61.3	–	–
Random split (sanity)	540 / 540	0.18	–	–
Within-subject (avg, 5 subjects)	*varies*	38.1	–	–

4.2 Temporal Comparability: 31% Item Instability

The central finding of this paper is that a substantial fraction of MMLU items behave differently across model generations. Comparing models released in 2023 ($N = 1{,}080$) against models released in 2024 ($N = 807$), 31.3% of items receive ETS C classification ($|\Delta_{\mathrm{MH}}| \geq 1.5$). Applying Benjamini-Hochberg FDR correction at $q = 0.05$ reduces this from 31.32% to 31.30%, removing only 3 items. The minimal impact of FDR correction reflects the fact that the ETS classification system imposes a dual threshold on both effect size and statistical significance, which already provides implicit multiple-testing protection. In absolute terms, nearly one in three MMLU items exhibits a large shift in its statistical relationship to overall model ability between the two cohort periods.

Three control conditions establish that this instability reflects genuine temporal change rather than methodological artifacts. A random split of the 2023 cohort into two equal halves produces only 0.18% ETS C items, confirming that sampling noise alone does not generate large-scale DIF. An ability-quintile comparison (lowest-scoring vs. highest-scoring models within the same temporal cohort) yields 2.9% ETS C items, indicating that ability differences alone are insufficient to produce the observed instability. A within-family comparison of Llama-2 ($N = 161$) and Llama-3 ($N = 308$) models produces 61.3% ETS C items. This elevated rate is consistent with the known sensitivity of MH-DIF to ability differences between groups (Holland & Thayer, 1988): models from different generations within a single family exhibit large ability gaps that confound the DIF signal. The contrast between 2.9% (ability-controlled) and 61.3% (ability-confounded) brackets the temporal rate of 31.3% and confirms that it captures a mixture of genuine item instability and residual ability differences.

The instability penetrates within every knowledge domain. Within-subject analysis yields an average of 38.1% ETS C items across all five MMLU subjects (range: 31–45%), and the ecological correlation between subject-level accuracy improvement and subject-level C% is $\rho = 0.307$ ($p = 0.02$). This ecological correlation could suggest that domains where models improved most also exhibit the most instability. Item-level logistic regression, however, reveals that domain accuracy improvement explains only 2.4 percentage points of the 31.3% temporal C% (pseudo $R^2 = 0.0007$; counterfactual C% with domain improvement regressed out $= 29.0\%$). Of the observed instability, 92.4% is unrelated to aggregate domain-level accuracy gains and instead reflects item-specific changes in how models interact with individual questions. After controlling for item difficulty and discrimination (pseudo $R^2 = 0.042$), the coefficient of domain improvement reverses sign (Simpson's paradox), further confirming that domain-level accuracy gains do not drive the observed instability.

Architecture heterogeneity is a non-issue for this analysis. The METABENCH MMLU population is 99.98% decoder-only (5,226 of 5,227 models). Excluding the single encoder-decoder model yields C% $= 31.32\%$ with Jaccard similarity of 1.0 between flagged item sets. Threshold sensitivity analysis confirms robustness across the full ETS range: C% $= 48.9\%$ at $|\Delta_{\mathrm{MH}}| \geq 1.0$, 31.3% at ≥ 1.5 (the standard ETS C threshold), 18.3% at ≥ 2.0, and 10.2% at ≥ 2.5. The qualitative conclusion holds at every threshold. Table 1 summarizes the DIF landscape across all cohort comparisons.

4.3 Ground Truth Mismatch

If DIF captures differential contamination, flagged items should overlap with independently identified contaminated items. We evaluate this hypothesis against the web-overlap labels of Li et al. (2024b),

Table 2: Structural approaches to reducing temporal DIF. None reduces baseline FPR below 31%, indicating that the observed instability is not an artifact of ability matching. Subscript $\pm$ values denote standard deviations across 5 injection seeds (baseline row only).

Approach	Mechanism	Temporal FPR	TPR (p=0.5)	Outcome
Baseline (external)	Cross-benchmark total score	$0.314_{\pm.009}$	$0.834_{\pm.027}$	Reference
Matched-ability	Within-quintile temporal DIF	$\sim$0.31	–	Enrich. 0.94–1.04
θ-conditioning	θ replaces total score	0.312	0.876	$r(\theta, \text{total})$=0.9986
Within-subject	Per-subject DIF	–	–	C% $= 38.1\%$ (>25%)

which classify MMLU items as "contaminated" based on verbatim matches in web training corpora. The enrichment ratio of DIF flags against these labels is 0.96 (Table 1), indistinguishable from chance. This null result does not indicate that DIF fails to detect contamination. Semi-synthetic injection experiments demonstrate a clear dose-response relationship: injecting controlled differential contamination at exploitation rate $p = 0.5$ produces $\Delta\text{TPR} = +0.508$, and at $p = 1.0$, $\Delta\text{TPR} = +0.674$ (Figure 2). The disconnect between DIF flags and web-overlap labels arises because the two methods answer fundamentally different questions.

Web-overlap labels measure *global exposure*: whether an item exists in publicly accessible training data. DIF measures *differential functioning*: whether an item's statistical relationship to overall ability changes across temporal cohorts. An item can be globally contaminated yet function identically across cohorts if all model generations have comparable access to the same training data. Conversely, an item can exhibit large DIF without any identifiable contamination source if changes in training methodology alter how models process certain question types. The Li et al. labels are therefore the wrong ground truth for evaluating a differential detection method: the near-chance enrichment is expected, not evidence of failure.

4.4 Structural Analysis

Three progressively refined approaches fail to reduce the temporal false positive rate below 31%, revealing a structural property of the data rather than a methodological limitation. Matched-ability quintile analysis restricts comparisons to models of similar total score and yields enrichment ratios of 0.94–1.04, with temporal FPR unchanged at approximately 0.31. IRT θ-conditioning with 20 ability strata achieves $r(\theta, \text{total score}) = 0.9986$ and FPR $= 0.312$, confirming that the external total score already captures nearly all ability variance relevant to DIF computation. Within-subject DIF analysis, intended to isolate subject-specific effects, finds C% $= 38.1\%$ across subjects, exceeding the 25% threshold that would indicate viable bifactor structure. Table 2 summarizes these results. The temporal FPR reflects genuine item-level instability that persists regardless of how ability is measured or conditioned. Yen's Q3 diagnostic reveals mild local dependence within subjects (11.3% of within-subject item pairs exceed $|Q3| > 0.20$, vs. 2.2% cross-subject; $5\times$ ratio, $p = 2.3 \times 10^{-9}$), which contributes to elevated within-subject DIF rates but is not the primary driver: the cross-subject matching protocol addresses this bias, and temporal C% persists under external matching (full Q3 distributions in Appendix Figure A2).

4.5 Semi-Synthetic Framework Validation

The semi-synthetic injection framework enables controlled evaluation of DIF detection by flipping incorrect responses to correct for a designated "contaminated" subset of items in the focal cohort at a specified exploitation rate p. The critical methodological finding is that the choice of matching strategy determines whether the framework produces valid results. Cross-subject matching computes the matching variable from subjects *other than* the one being tested, ensuring that the conditioning variable is unaffected by the injected contamination. Under this protocol, $\Delta\text{FPR} = 0.000$ across all five MMLU subjects, and four of five subjects produce $\Delta\text{TPR} > 0.3$ at $p = 0.5$ (Figure 3). Within-subject matching, by contrast, computes the matching variable from the same subject where contamination was injected, which inflates FPR by an average of $+0.186$ because the matching variable absorbs part of the injection signal and violates the conditional independence assumption of the MH procedure. We adopt cross-subject matching as the standard protocol for all subsequent analyses and recommend it as default for semi-synthetic DIF evaluation on multi-subject benchmarks.

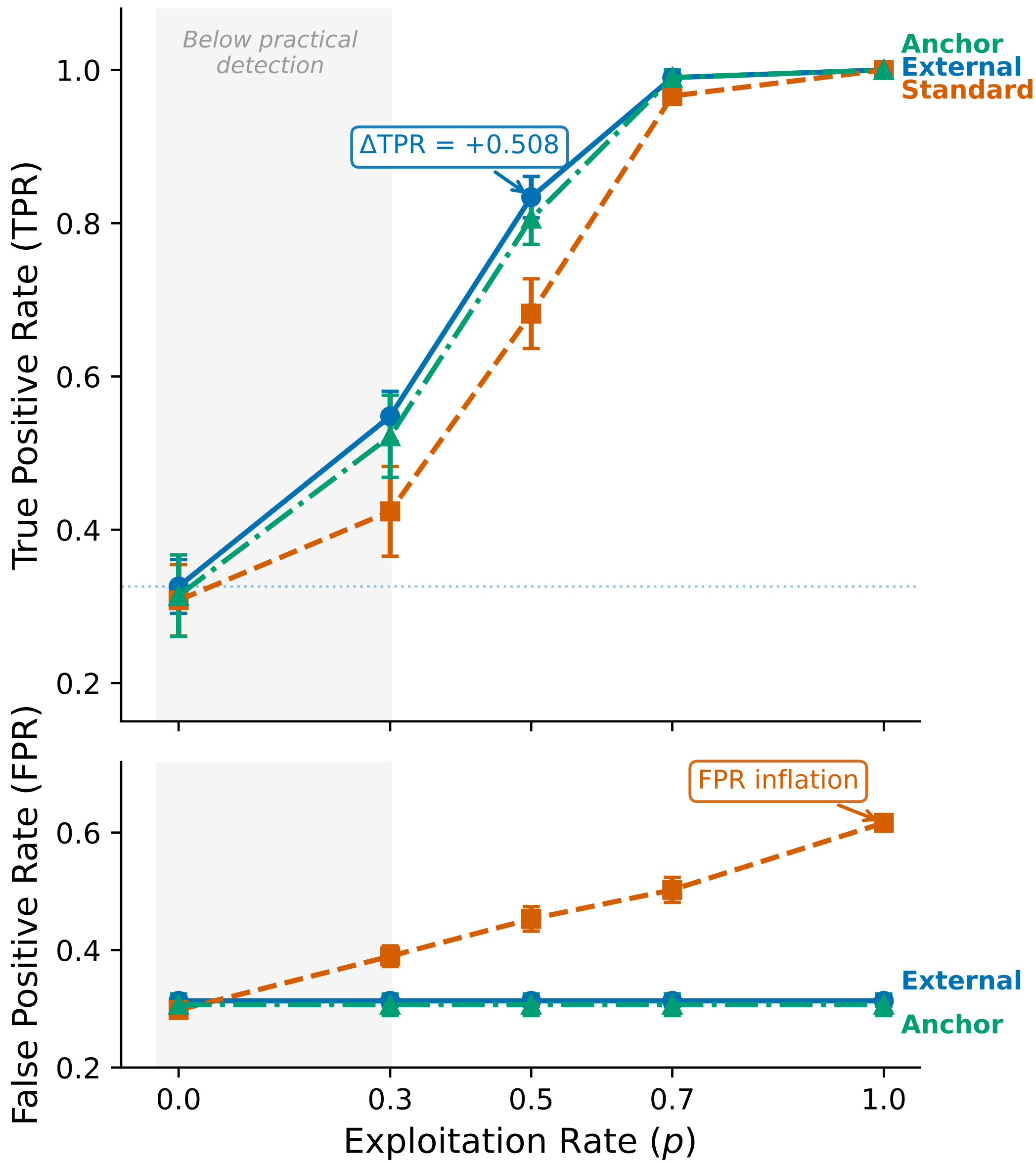

Figure 2: Semi-synthetic dose-response for DIF detection on MMLU. ΔTPR increases monotonically with exploitation rate p, confirming that MH-DIF detects differential contamination when present. Baseline (no injection) produces $\Delta\text{TPR} \approx 0$.

4.6 Downstream Impact

The practical relevance of 31% item instability depends on whether unstable items affect model evaluation and benchmark rankings. We investigate this through a four-layer analysis that progresses from aggregate statistics to model-specific diagnostics.

Layer 1: Benchmark redundancy. Spearman correlation between model rankings computed on all 12,508 items versus stable-only items (ETS A/B) is $\rho \approx 0.997$. At the level of aggregate rankings, removing 31% of items changes almost nothing. This redundancy is a consequence of the large item pool: 12,508 items provide sufficient statistical mass that removing nearly a third does not alter rank order.

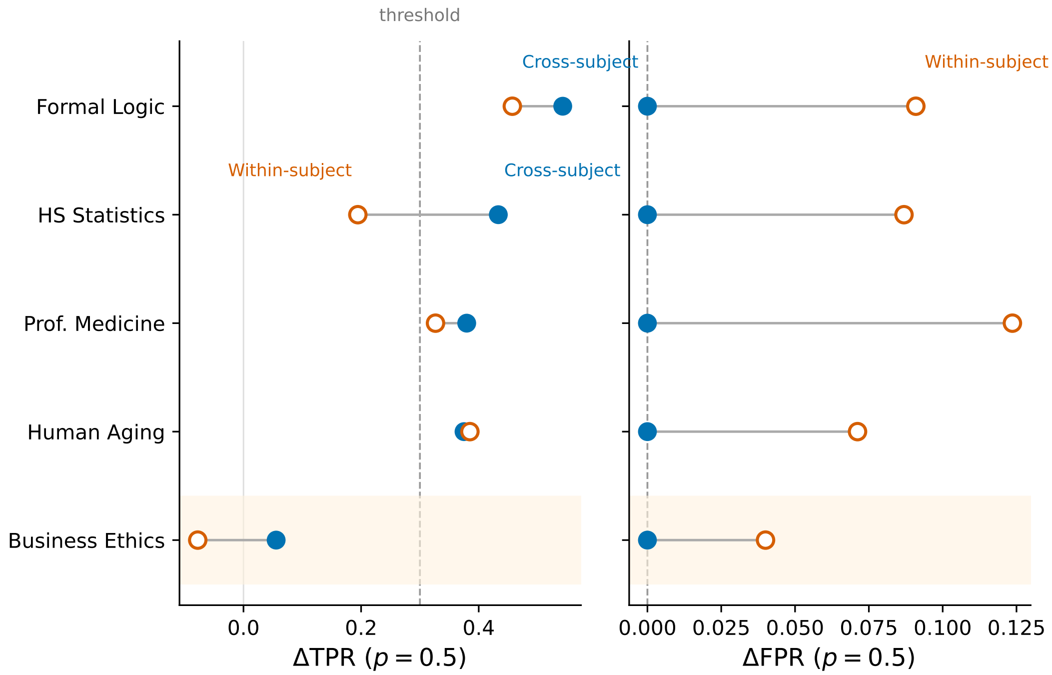

Figure 3: Cross-subject vs. within-subject matching for semi-synthetic DIF detection. Cross-subject matching eliminates FPR inflation ($\Delta\text{FPR} = 0.000$) while preserving detection power ($\Delta\text{TPR} > 0.3$ for 4/5 subjects at $p = 0.5$). Within-subject matching inflates FPR by $+0.186$ on average.

Layer 2: Signal concentration with circularity check. Despite aggregate redundancy, DIF-flagged items carry more discriminative signal than the average item. Removing DIF items degrades ranking discriminability more than removing an equal number of random items ($z = -8$ to -49 across 21 experimental conditions). A circularity ablation, however, places this in context: DIF-based removal is the *least* disruptive among non-random removal strategies. Accuracy-gap removal produces $z = -156.5$, high-variance removal $z = -107.9$, and high-discrimination removal $z = -15.8$, compared to $z = -9.5$ for DIF removal. The Jaccard overlap between DIF-flagged and high-discrimination items is only 0.12. DIF items carry more signal than average, but DIF isolates differential behavior across cohorts specifically, not generally informative items.

Layer 3: Diagnostic lens. Individual model families show distinctive DIF profiles that reveal which families disproportionately benefit from unstable items. Phi-3 and Llama-3 models gain $+700$ to $+850$ rank positions when evaluated on stable items only, suggesting inflated performance on items whose function changed between 2023 and 2024. Mistral-based model merges show the opposite pattern, losing -400 to -587 positions. A caveat applies: among the top 50 rank movers, most lack documented training details in METABENCH, limiting interpretation of what training choices drive these rank shifts.

Layer 4: Cross-benchmark signal. The DIF instability pattern extends beyond MMLU. Section 4.7 presents a full replication on GSM8K with independent contamination validation using the GSM1K held-out set (Zhang, 2024). Figure 4 visualizes the relationship between full and DIF-filtered rankings.

4.7 Cross-Benchmark Validation: GSM8K

To test whether temporal DIF is specific to MMLU or reflects a broader phenomenon in LLM benchmarking, we replicate the full analysis pipeline on GSM8K (1,319 items × 6,702 models from METABENCH).

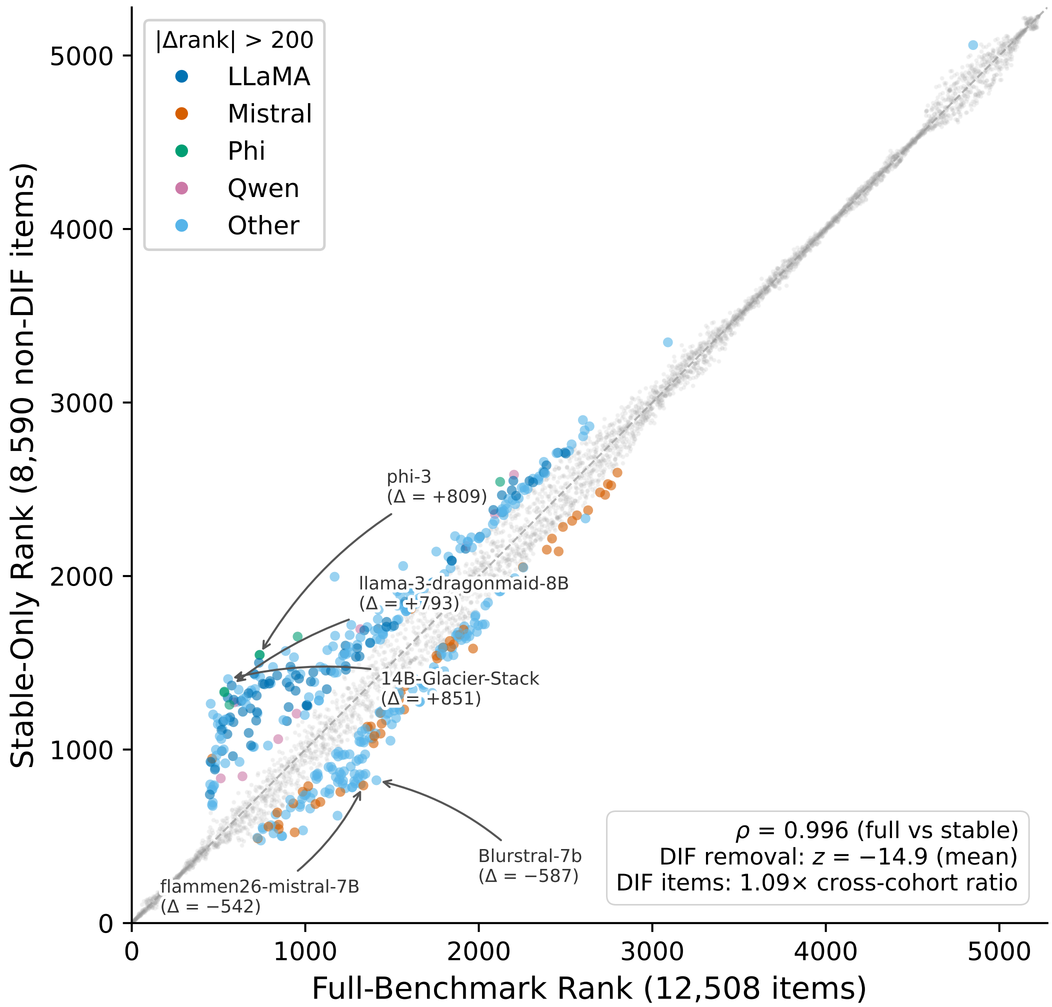

Figure 4: Downstream impact of DIF-based item filtering on model rankings. Despite $\rho \approx 0.997$ overall, individual model families show systematic rank shifts when evaluated on stable items only, with Phi-3 and Llama-3 gaining and Mistral merges losing rank positions.

Temporal DIF. Standard matching yields $\mathrm{C\%} = 18.9\%$ for the temporal cohort comparison, with a random-split sanity check at 0.3%. The lower instability rate compared to MMLU (31.3%) is consistent with GSM8K's more homogeneous item structure: single-domain mathematical reasoning versus MMLU's 57 diverse subjects. The sanity check confirms that the signal is non-random.

Semi-synthetic validation. Dose-response confirms detection power on this benchmark. Standard matching achieves $\Delta\mathrm{TPR} = +0.818$ at exploitation rate $p = 0.7$. External matching (using scores on ARC, HellaSwag, TruthfulQA, and Winogrande to match models) yields $\Delta\mathrm{TPR} = +0.514$ at $p = 0.5$, comparable to MMLU's $+0.508$ at the same exploitation rate, with $\Delta\mathrm{FPR}$ well-controlled. Table 3 compares key metrics across both benchmarks.

GSM1K contamination gap. The GSM1K dataset (Zhang, 2024) provides an independent measure of per-model contamination by comparing performance on the original GSM8K versus a held-out set of novel problems. For $N = 33$ models with both GSM1K scores and METABENCH evaluations, we correlate each model's contamination gap (GSM8K accuracy minus GSM1K accuracy) with the ranking advantage gained from DIF-flagged items. The observed correlation is $r = -0.478$ (95% CI $= [-0.71, -0.16]$, $p = 0.005$; item-permutation $p = 0.014$), with partial $r = -0.385$ ($p = 0.027$) after controlling for overall accuracy. Our original hypothesis predicted a *positive* correlation: models

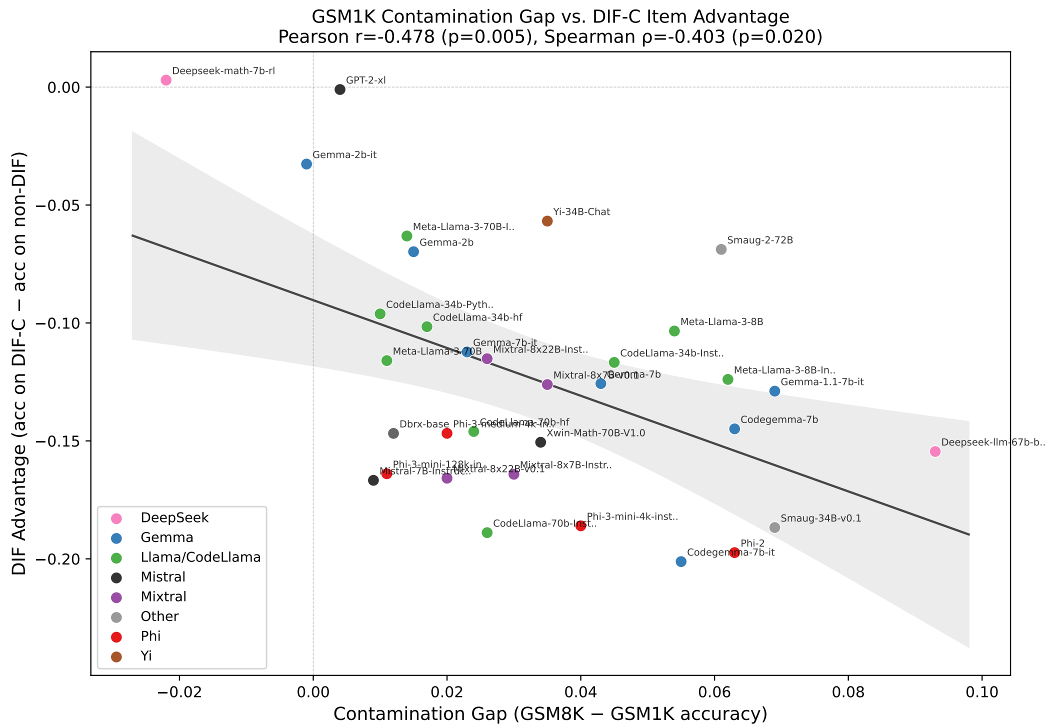

Figure 5: GSM1K contamination gap vs. DIF ranking advantage for $N = 33$ models. The negative correlation ($r = -0.478$, $p = 0.005$) indicates that models with larger contamination gaps benefit less from DIF-flagged items, contrary to the initial positive-correlation hypothesis.

Table 3: Cross-benchmark comparison of DIF analysis on MMLU and GSM8K. Both benchmarks exhibit temporal item instability with validated semi-synthetic detection power. Standard and external matching use different exploitation rates (p) because each benchmark's optimal operating point differs; external matching results are compared at $p = 0.5$ for both. Subscript $\pm$ values denote standard deviations across 5 random injection seeds.

Metric	MMLU	GSM8K
Items	12,508	1,319
Models	5,227	6,702
Temporal C%	31.3%	18.9%
Sanity C%	0.18%	0.3%
ΔTPR (p=0.7, standard)	–	$+0.818_{\pm.006}$
ΔTPR (p=0.5, external)	$+0.508_{\pm.027}$	$+0.514_{\pm.028}$

benefiting more from contamination should also benefit more from DIF items. The observed *negative* correlation indicates the opposite. DIF-C items resist contamination-based performance gains, and models with larger contamination gaps benefit *less* from these items. This result is suggestive at $N = 33$ and warrants replication with larger model populations, but it aligns with the broader interpretation that DIF captures item-level behavioral change distinct from simple memorization of training data. Figure 5 shows the correlation scatter.

5 Discussion

5.1 Implications for Benchmark Temporal Comparability

The 31% ETS C instability rate is robust across thresholds (10% at $|\Delta_{\mathrm{MH}}| \geq 2.5$, rising to 49% at ≥ 1.0), and three structural controls (Section 4.4) confirm it is not a methodological artifact.

This means that cross-generation MMLU score comparisons carry uncontrolled item-level variance: nearly one in three items functions differently depending on when the evaluated model was trained. The instability decomposition (Section 4.2) sharpens this interpretation. Domain-level accuracy improvement explains only 2.4 percentage points of the 31.3% C% (pseudo $R^2 = 0.0007$), so 92.4% of the instability is unrelated to aggregate capability gains. The items that change are not simply the ones that newer models answer correctly because they are better at a domain; instead, they change in item-specific ways that domain-level trends do not predict.

This instability does not invalidate MMLU as a ranking instrument. Spearman $\rho \approx 0.997$ between full and stable-item rankings confirms that aggregate model ordering is preserved. The instability is diagnostic, not destructive: it identifies which measurement units shift across model generations without rendering the overall score unreliable. Benchmark designers should therefore consider monitoring DIF as a quality diagnostic alongside accuracy metrics. An item pool where C% grows over successive model cohorts signals that the benchmark's measurement properties are drifting, even if rank ordering remains intact.

5.2 Ground Truth Meta-Critique

The near-chance enrichment (0.96) between DIF flags and web-overlap labels raises a question that extends beyond our specific analysis. All prior work that validates item-level contamination detectors against web-overlap or n-gram labels faces the same methodological gap: these labels measure whether an item exists in publicly accessible training corpora, not whether exposure differentially affects model behavior. Global exposure is a property of the data ecosystem; differential functioning is a property of how model cohorts interact with individual items. Most MMLU items are globally contaminated in the sense that they appear in web text, so web-overlap labels identify a condition shared by nearly all items rather than one that differentiates model cohorts. Validating differential detectors against such labels conflates two distinct phenomena.

This observation constitutes a meta-contribution to the contamination detection literature. The enrichment ratio of ≈ 1.0 is not evidence that DIF fails; it is the expected outcome when a differential method is evaluated against a global-exposure reference. Semi-synthetic injection confirms that DIF detects differential contamination when present ($\Delta\text{TPR} = +0.508$ at $p = 0.5$; Section 4.5). The field needs validation methodologies that match the granularity of the detection task: item-level differential detectors require item-level differential ground truth, not item-level global-exposure labels.

5.3 DIF as Diagnostic Lens

Capability diagnostic. The GSM1K correlation ($r = -0.478$, $p = 0.005$, $N = 33$; Section 4.7) provides evidence that DIF-flagged items capture behavioral change distinct from memorization. Models with larger contamination gaps (higher GSM8K minus GSM1K accuracy) benefit *less* from DIF-C items, the opposite of what a pure memorization account would predict. Of the 249 focal-favoring DIF-C items, 171 have discrimination $\alpha < 1$, and the negative correlation concentrates in this low-discrimination subset. This pattern suggests that contamination produces a "shallow" performance boost on items that DIF does not flag, while DIF-C items capture genuine capability differences that contamination does not confer. Partial correlation ($r = -0.385$, $p = 0.027$) after controlling for overall accuracy confirms that the effect is not an artifact of general model quality. Several caveats apply: the confidence interval is wide (95% CI = $[-0.71, -0.16]$), item difficulty as a confound is not fully excluded, and $N = 33$ limits generalizability. This result is suggestive and warrants replication with larger model populations.

Structural constraints on DIF applicability. Systematic experimentation reveals three structural constraints that jointly limit DIF utility in LLM benchmarks. First, 1D IRT model fit is imperfect: MMLU's multi-domain structure produces mild local dependence (11.3% of within-subject item pairs exceed $|Q3| > 0.20$; Section 4.4), which elevates within-subject DIF rates relative to cross-subject baselines. Second, the power analysis (Section 4.1) establishes minimum requirements of $N \geq 80$ per cohort and $|\Delta_{\text{MH}}| \geq 1.4$ for reliable detection. Benchmarks with small model populations or modest effect sizes will lack statistical power for DIF analysis. Third, ability differences between temporal cohorts confound the DIF signal: the within-family comparison (Llama-2 vs. Llama-3) produces 61.3% C%, elevated above the 31.3% temporal rate by the large ability gap between model generations. These are structural constraints revealed by our experimentation, not fundamental impossibilities. Higher-level methods (MIMIC models, propensity-score matching, multidimensional IRT) may address individual constraints, and their investigation on LLM benchmark data is a natural next step.

5.4 LIMITATIONS

(a) Ecological vs. item-level decomposition. The ecological correlation between subject-level C% and accuracy improvement is $\rho = 0.307$ ($p = 0.02$), but item-level decomposition shows that domain improvement explains only 2.4 percentage points (7.6%) of 31.3% C%. The remaining 92.4% cannot be fully attributed to contamination; other sources of capability change (training methodology, data composition, optimization procedures) may contribute, and our analysis does not isolate these factors.

(b) Instability attribution. The 31% C% measures items that function differently across cohorts, but the method does not distinguish contamination from legitimate capability change. DIF identifies *that* items changed, not *why*. The semi-synthetic framework validates detection power for one specific mechanism (differential exposure), but real-world instability likely reflects multiple interacting causes.

(c) Semi-synthetic injection fidelity. The injection procedure flips incorrect responses to correct at a specified rate, simulating an idealized contamination scenario. Real contamination may manifest as partial knowledge, format memorization, or selective retrieval rather than wholesale answer flipping. The gap between our injection model and realistic contamination mechanisms is unknown.

(d) Benchmark scope. The primary analysis covers MMLU with cross-benchmark validation on GSM8K (Section 4.7). GSM8K replication confirms framework portability and provides the GSM1K contamination correlation ($N = 33$), but full temporal DIF characterization on diverse benchmarks (code generation, long-form reasoning, multimodal tasks) has not been conducted. The GSM1K sample ($N = 33$) produces wide confidence intervals that limit the precision of the negative correlation estimate.

(e) Cross-subject baseline false positive rate. The cross-subject matching protocol achieves $\Delta\mathrm{FPR} = 0.000$ for injected contamination but does not reduce the baseline temporal FPR of 33–53% across subjects. This rate is stable across injection levels, confirming it reflects genuine temporal instability rather than methodological error, but it means that a large fraction of flagged items in temporal comparisons are not attributable to the specific contamination mechanism under test.

(f) RL model robustness. Reinforcement learning from human feedback may alter item-level response patterns in ways that differ from pretraining-based contamination. The theoretical motivation for DIF robustness to RL (RL modifies the ability distribution rather than item parameters) is plausible but empirically unvalidated with adequate sample size: METABENCH contains only $N = 13$ models with documented reasoning-RL training.

(g) Model family coverage. Downstream impact analysis is limited to families with $N \geq 20$ models (Llama, Mistral, Qwen, Phi). Among the top 50 rank movers, 49 of 50 belong to the "unknown" family category in METABENCH, precluding interpretation of what training choices produce large rank shifts.

(h) Future directions. Cross-benchmark generalization to code, multimodal, and long-form reasoning benchmarks would test DIF applicability beyond factual knowledge tasks. RL model cohorts with adequate N would enable empirical validation of the RL robustness hypothesis. MIMIC models and propensity-score matching could address the ability confounding that the current MH-DIF framework does not fully resolve.

6 Conclusion

Applying Mantel-Haenszel DIF analysis to the METABENCH response matrix (5,227 models $\times$ 12,508 MMLU items), we find that 31% of items are functionally unstable across model generations under ETS Category C classification. The instability is robust across thresholds (10–49% at $|\Delta_{\mathrm{MH}}| \in [2.5, 1.0]$), confirmed non-random by a random-split control (0.18%), and predominantly item-specific: domain accuracy improvement accounts for only 2.4 percentage points of the 31.3% C%. This instability does not invalidate MMLU as a ranking instrument ($\rho \approx 0.997$ between full and stable-item rankings) but signals that item-level measurement properties are drifting across model generations.

The analysis exposes a validation methodology gap in the contamination detection literature. Web-overlap labels measure whether items exist in publicly accessible training corpora, not whether items function differently across model cohorts. The enrichment between DIF flags and these labels is 0.96 (chance level), the expected outcome when a differential method is evaluated against global-exposure labels. Our semi-synthetic injection framework with cross-subject matching ($\Delta\mathrm{FPR} = 0.000$, $\Delta\mathrm{TPR} = +0.508$ at $p = 0.5$) provides a controlled alternative that any item-level detector can use for validation. Cross-benchmark replication on GSM8K ($\Delta\mathrm{TPR} = +0.818$) confirms the framework's portability.

DIF-flagged items are not noise. They concentrate cross-generational discriminative information ($z = -8$ to -49σ vs. random removal) and anti-correlate with the GSM1K contamination gap ($r = -0.478$, $p = 0.005$, $N = 33$), consistent with DIF items capturing genuine capability change that contamination does not confer. Individual model families show distinctive DIF profiles: Phi-3 and Llama-3 gain $+700$ to $+850$ rank positions under stable-item evaluation, while Mistral-based merges lose -400 to -587 positions. DIF is a diagnostic lens for understanding what changed across model generations, not a quality filter for removing bad items.

LLM benchmarks need psychometric monitoring alongside score reporting. DIF analysis provides a scalable, method-agnostic diagnostic that requires only a binary response matrix and group labels, with no access to model internals or training data.

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